

This study demonstrates that a granular understanding of CTCL and, more broadly, lymphoproliferative disorders have the potential to significantly improve clinical outcomes. Many questions remain unresolved. Addressing these will require dedicated investigations in large patient cohorts to validate the longitudinal findings across therapeutic settings. In particular, it will be imperative to define the contribution of host factors, the tissue environment, and the mechanisms controlling CTCL homing and its dynamic evolution across different compartments. Achieving these goals will require multi-institutional and multidisciplinary collaboration and sustained funding.

Conflict-of-interest disclosure: The authors declare no competing financial interests. ■

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RED CELLS, IRON, AND ERYTHROPOIESIS

Comment on [Sohail et al](#), page 2518

No need to wait: treat iron deficiency in the hospital

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In this issue of *Blood*, Sohail et al¹ seek to answer whether it is safe to administer IV iron therapy in an iron-deficient person who is hospitalized with an active infection. Their work, which assessed deidentified patient data across multiple decades, included individuals with pneumonia, cellulitis, methicillin-resistant *Staphylococcus aureus* bacteremia, urinary tract infection, or colitis. Across infection types, those who received IV iron therapy had higher rates of overall survival at both 14 and 90 days compared with individuals who did not, contradicting the notion that IV iron results in worse outcomes in the setting of infection.

Iron is a critical micronutrient that has key roles in countless cellular processes, including DNA replication, neurotransmitter synthesis, and mitochondrial energy production.² As hematologists, we are biased in believing that iron's role in erythropoiesis is the most important, and although we certainly need sufficient iron to prevent anemia and maximize oxygen-carrying capacity, it is also essential for optimal immune function and overall health. An iron-deficient state not only increases the risk of anemia but also affects individuals' ability to fully recover from acute illness and hospitalization.

Despite iron's vital roles, iron deficiency is common. In the general adult population, the global prevalence of iron deficiency is 16%.³ Rates are higher for those individuals who are menstruating (27%) or pregnant (37% or more).⁴ Furthermore, ~40% of critically ill patients have anemia at the time of hospital admission, but this number doubles to 80% at the time of discharge.⁵ In the hospital setting, anemia is associated with longer length of stay and higher morbidity and mortality.^{5,6} Iron deficiency contributes to more than half of anemia cases in hospitalized patients, with inflammation being another common cause. It stands to reason that addressing anemia in hospitalized patients should improve outcomes. In

patients with concomitant iron deficiency and inflammation, this means consideration of IV iron therapy.

Alas, the administration of iron in the setting of infection has long been controversial and not without good reason. Iron is necessary not only for humans but for essentially all organisms, including parasites, bacteria, and viruses.⁷ Certain infectious states, such as malaria or involving siderophilic bacteria, do worsen in the setting of iron supplementation. Furthermore, as part of an immunoprotective response, inflammation upregulates hepcidin, which results in iron sequestration within the body, thereby reducing iron availability to pathogens.⁷ As a result of this knowledge, many clinicians avoid IV iron in the setting of infection due to concerns about feeding pathogens and potentially causing harm to the patient.

These same principles may not hold true, however, for other forms of infections such as pneumonia or cellulitis, and existing literature looking at such outcomes is mixed. A 2015 review on the overall safety of IV iron preparations compared with oral iron found comparable severe adverse events, with no increased risk of infections.⁸ A 2021 systematic review and meta-analysis evaluating the risk of IV iron compared

with oral iron or no iron suggested an increased risk of infection (risk ratio [RR], 1.17) but also demonstrated an increase in hemoglobin (0.57 g/dL), a reduction in red blood cell transfusion (RR, 0.93), and, importantly, no effect on mortality or hospital length of stay.⁹

Thus, the following question has persisted: In hospitalized patients with active infection and iron deficiency anemia, does IV iron therapy worsen infection or improve patient outcomes? The study by Sohail et al sought to address this question by evaluating outcomes in hospitalized adults with acute infections and iron deficiency anemia who did or did not receive IV iron therapy using data from a large research network over a time span of 25 years. Using a large electronic health database of >100 health care organizations, the authors identified all patients 18 years of age or older admitted with frequently encountered acute infections in the inpatient setting. Using propensity matching, they identified thousands of patients in each infection group who had or had not received IV iron therapy for inclusion in the final analysis. Patients who received IV iron had better survival, greater improvements in hemoglobin, and decreased receipt of red cell transfusions, with no change in overall length of hospital stay.¹

Consulting hematologists are often asked about the treatment of anemia in the hospital setting. These findings provide real-world data that associate IV iron therapy with improved survival. Significantly, it did not demonstrate an increased risk of harm. Although the data are impressive, they are retrospective in nature and the clinical decision-making around administration of IV iron therapy cannot be discerned. Furthermore, as the data were not derived from a clinical trial, we can only conclude that IV iron is associated with benefits rather than directly causative of them. The data do, however, support the conduct of randomized trials of IV iron therapy that are inclusive, rather than exclusive, of patients with infections. Including these patient groups would allow us to evaluate these outcomes more rigorously and confirm safety and benefits. Knowing that a readily available treatment for iron deficiency improves survival has the potential for tremendous impact on hospital practice and should spur initiatives

to appropriately diagnose and treat as many affected patients as possible.

Conflict-of-interest disclosure: The author declares no competing financial interests. ■

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THROMBOSIS AND HEMOSTASIS

Comment on [Maroney et al](#), page 2530

No TFPI? APC to the rescue!

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In this issue of *Blood*, Maroney et al¹ document the ability of a hyper-activatable variant of mouse protein C to partially rescue the embryonic lethality and developmental abnormalities associated with germ line deficiency of tissue factor pathway inhibitor (TFPI) in mice.

Regulated clotting system activity is critical for maintaining hemostasis and promoting proper healing after injury. However, over the last several decades, studies have revealed that genetic deficiencies in coagulation factors (eg, prothrombin, factor X [FX], tissue factor) result in perturbation of vascular development and embryonic lethality with extensive bleeding, leading to the concept of developmental hemostasis.² Intriguingly, germ line deficiencies in genes encoding the anticoagulant proteins also result in embryonic lethality, characterized by organ fibrin and platelet deposition but also bleeding.² In 1997, Huang et al generated TFPI mutant mice (*Tfpi*^{-/-})

deficient in the first Kunitz domain. *Tfpi*^{-/-} mice, which uniformly die during embryogenesis starting around day E9.5 with yolk sac hemorrhage but some embryos survive up to birth.³ Those embryos that survive to E12.5 and beyond develop vascular malformations termed glomeruloid bodies and fibrin deposition specifically in the brain, but not in other tissues.⁴

The primary function of TFPI is inhibition of the tissue factor/activated FVII complex. However, numerous studies suggest that TFPI appears to be particularly important in suppressing the activity of FV collaboratively with the thrombomodulin (Thbd)/activated protein C (aPC) pathway.